HANCHENG MIN

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Institute of Natural Sciences & School of Mathematical Sciences
Shanghai Jiao Tong University

EDUCATION

Johns Hopkins University, Baltimore, MD, U.S.A. *Sep. 2018 – Jul. 2023*
Ph.D., Electrical and Computer Engineering

University of Pennsylvania, Philadelphia, PA, U.S.A. *Sep. 2016 – May. 2018*
Master of Science in Engineering, Electrical and Systems Engineering

Tongji University, Shanghai, China *Sep. 2012 – Jul. 2016*
Bachelor of Engineering, Automation

ACADEMIC APPOINTMENTS

Tenure-track Associate Professor *Sep. 2025 – Present*
Institute of Natural Sciences and School of Mathematical Sciences
Shanghai Jiao Tong University

Postdoctoral Researcher *Aug. 2023 – Jul. 2025*
Vidal-lab, Electrical and Systems Engineering, *Advisor*: René Vidal
University of Pennsylvania

Graduate Research Assistant *Sep. 2018 – Jul. 2023*
NetD-lab, Electrical and Computer Engineering, *Advisor*: Enrique Mallada, René Vidal
Johns Hopkins University

Graduate Research Assistant *Jun. 2017 – May. 2018*
Kod*lab, Electrical and Systems Engineering, *Mentor*: Ömür Arslan
University of Pennsylvania

PROJECTS

Grant awarded

- **SJTU** Traing Dynamics of Generative Models from a Feature Learning Perspective, 2027-2028
- **NSFC** Convergence Analysis of Muon Optimizer from a Feature Learning Perspective, 2027-2029
- **Science and Technology Commission of Shanghai Municipality** Theory of Neural Collapse in Deep Neural Networks, 2026-2027
- **NSFC** Theory of Latent Representation in Deep Learning with Applications, 2026-2028
- **SJTU** Institute of Natural Sciences Startup Fund, 2025

Grant as contributor

- **NSF-Simons** Collaborative Research: Transferable, Hierarchical, Expressive, Optimal, Robust, Interpretable Networks, 2021-2024
- **NSF** HDR TRIPODS: Institute for the Foundations of Graph and Deep Learning, 2021-2023
- **ONR** MURI: Control and Learning Enabled Verifiable Robust AI, 2021-2025

AWARDS AND HONORS

AI x Science Postdoctoral Fellowship, Data Driven Discovery Initiative at Penn *Nov. 2024*
MINDS Data Science Spring Fellowship 2021 *Jan. 2021*
MINDS Data Science Fellowship 2019/2020 *Nov. 2019*
ICRA 2018 Best Paper in Multi-Robot Systems (nominee) *Mar. 2018*
Tongji Scholarship of Excellence *2013-2015*
Chinese Mathematics Competition (Shanghai Preliminary) *Nov. 2013*

PUBLICATIONS

Preprints

[P1] **H. Min**, S. Tarmoun, R. Vidal, and E. Mallada, "Convergence and implicit bias of gradient flow on overparametrized linear networks," 2023, arXiv:2105.06351.

Journals

[J6] **H. Min**, L. E. MacDonald, and R. Vidal, "On the convergence, implicit bias and edge of stability of gradient descent in deep learning," *IEEE Signal Processing Magazine (**IEEE SPM**)*, vol. 43, no. 3, pp. 37–51, 2026.

[J5] **H. Min**, R. Pates, and E. Mallada, "A frequency domain analysis of slow coherency in networked systems," ***Automatica***, vol. 174, p. 112 184, 2025.

[J4] Z. Xu, **H. Min**, S. Tarmoun, E. Mallada, and R. Vidal, "A local Polyak-Lojasiewicz and descent lemma of gradient descent for overparameterized linear models," *Transactions on Machine Learning Research (**TMLR**)*, 2025.

[J3] Y. Jiang, **H. Min**, and B. Zhang, "Oscillations-aware frequency security assessment via efficient worst-case frequency Nadir computation," *Electric Power Systems Research (**EPSR**)*, vol. 234, pp. 110 656–110 664, 2024, also in PSCC 2024.

[J2] A. Castellano, **H. Min**, J. A. Bazerque, and E. Mallada, "Learning to act safely with limited exposure and almost sure certainty," *IEEE Transactions on Automatic Control (**IEEE TAC**)*, vol. 68, no. 5, pp. 2979–2994, May 2023.

[J1] **H. Min**, F. Paganini, and E. Mallada, "Accurate reduced order models for coherent heterogeneous generators," *IEEE Control Systems Letters (**IEEE L-CSS**)*, vol. 5, no. 5, pp. 1741–1746, Nov. 2021, also in ACC 2021.

Conferences

[C24] F. Tian, J. Luo, U. K. R. Tadipatri, **H. Min**, and R. Vidal, "Dynamic world generation made efficient," in *European Conference on Computer Vision (**ECCV**)*, Sep. 2026.

[C23] H. Li, **H. Min**, and R. Vidal, "Transformers learn the optimal DDPM denoiser for multi-token GMMs," in *International Conference on Machine Learning (**ICML**)*, Jul. 2026.

[C22] Z. Xu, **H. Min**, L. E. MacDonald, S. Tarmoun, E. Mallada, W. Su, and R. Vidal, "A margin perspective on LoRA: Robustness to catastrophic forgetting and adapter merging (MaLoRA)," in *Conference on Neural Information Processing Systems (**NeurIPS**)*, to appear, 2026.

[C21] L. E. MacDonald, L. Palma, Z. Xu, **H. Min**, S. Tarmoun, and R. Vidal, "Convergence rates for gradient descent on the edge of stability for overparametrised least squares," in *Conference on Neural Information Processing Systems (**NeurIPS**)*, Dec. 2025.

[C20] **H. Min** and R. Vidal, "Understanding incremental learning with closed-form solution to gradient flow on overparametrized matrix factorization," in *IEEE Conference on Decision and Control (**CDC**)*, Dec. 2025.

[C19] **H. Min**, Z. Zhu, and R. Vidal, "Neural collapse under gradient flow on shallow ReLU networks for orthogonally separable data," in *Conference on Neural Information Processing Systems (**NeurIPS**)*, Dec. 2025.

[C18] F. Tian, T. Ding, J. Luo, **H. Min**, and R. Vidal, "Voyaging into perpetual dynamic scenes from a single view," in *IEEE/CVF International Conference on Computer Vision (**ICCV**)*, Oct. 2025.

[C17] **H. Min** and R. Vidal, "Gradient flow provably learns robust classifiers for orthonormal GMMs," in *International Conference on Machine Learning (**ICML**)*, Jul. 2025.

[C16] J. Luo, T. Ding, K. H. R. Chan, **H. Min**, C. Callison-Burch, and R. Vidal, "Concept Lancet: Image editing with compositional representation transplant," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition(**CVPR**)*, Jun. 2025.

[C15] Z. Xu, **H. Min**, J. Luo, S. Tarmoun, L. E. MacDonald, E. Mallada, and R. Vidal, "Understanding the learning dynamics of LoRA: A gradient flow perspective on low-rank adaptation in matrix factorization," in *International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, May 2025.

[C14] **H. Min** and R. Vidal, "Can implicit bias imply adversarial robustness?" In *International Conference on Machine Learning (**ICML**)*, Jul. 2024.

[C13] Y. Jiang, **H. Min**, and B. Zhang, "Oscillations-aware frequency security assessment via efficient worst-case frequency Nadir computation," in *Power Systems Computation Conference (**PSCC**)*, Jun. 2024.

[C12] **H. Min**, E. Mallada, and R. Vidal, "Early neuron alignment in two-layer ReLU networks with small initialization," in *International Conference on Learning Representations (**ICLR**)*, May 2024.

[C11] A. Castellano, **H. Min**, J. A. Bazerque, and E. Mallada, "Learning safety critics via a non-contractive binary Bellman operator," in *Asilomar Conference on Signals, Systems, and Computers (**ACSSC**)*, Nov. 2023.

[C10] **H. Min**, R. Vidal, and E. Mallada, "On the convergence of gradient flow on multi-layer linear models," in *International Conference on Machine Learning (**ICML**)*, Jul. 2023.

[C9] **H. Min** and E. Mallada, "Learning coherent clusters in weakly-connected network systems," in *Learning for Dynamics and Control Conference (**L4DC**)*, Jun. 2023.

[C8] **H. Min** and E. Mallada, "Spectral clustering and model reduction for weakly-connected coherent network systems," in *American Control Conference (**ACC**)*, May 2023.

[C7] Z. Xu, **H. Min**, S. Tarmoun, E. Mallada, and R. Vidal, "Linear convergence of gradient descent for finite width over-parametrized linear networks with general initialization," in *International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, Apr. 2023.

[C6] A. Castellano, **H. Min**, J. A. Bazerque, and E. Mallada, "Reinforcement learning with almost sure constraints," in *Learning for Dynamics and Control Conference (**L4DC**)*, Mar. 2022.

[C5] **H. Min**, S. Tarmoun, R. Vidal, and E. Mallada, "On the explicit role of initialization on the convergence and implicit bias of overparametrized linear networks," in *International Conference on Machine Learning (**ICML**)*, Jul. 2021.

[C4] **H. Min**, F. Paganini, and E. Mallada, "Accurate reduced-order models for heterogeneous coherent generators," in *American Control Conference (**ACC**)*, Jun. 2021.

[C3] **H. Min** and E. Mallada, "Dynamics concentration of tightly-connected large-scale networks," in *IEEE Conference on Decision and Control (**CDC**)*, Dec. 2019.

[C2] **H. Min**, F. Paganini, and E. Mallada, "Accurate reduced order models for coherent synchronous generators," in *Allerton Conference on Communication, Control, and Computing (**Allerton**)*, Sep. 2019.

[C1] O. Arslan, **H. Min**, and D. E. Koditschek, "Voronoi-based coverage control of pan/tilt/zoom camera networks," in *IEEE International Conference on Robotics and Automation (**ICRA**)*, May 2018.

Thesis

[T2] **H. Min**, "Exploiting structural properties in the analysis of high-dimensional dynamical systems," Ph.D. Thesis, Johns Hopkins University, 2023.

[T1] **H. Min**, "On balancing event and area coverage in mobile sensor networks," Master Thesis, University of Pennsylvania, 2018.

TEACHING

Instructor

- *Artificial Intelligence Basics* (Spring 2026, Spring 2027; undergrad), Shanghai Jiao Tong University
- *Mathematical Analysis (Honor) II: Recitation* (Spring 2026; undergrad), Shanghai Jiao Tong University
- *Mathematical Analysis (Honor) I: Recitation* (Fall 2025; undergrad), Shanghai Jiao Tong University

Teaching Assistant

- *Foundations of Reinforcement Learning* (Fall 2020, Fall 2021, Fall 2022; grad), Johns Hopkins University
- *Control Systems* (Spring 2022; undergrad), Johns Hopkins University
- *Networked Dynamical Systems*, (Fall 2019; grad), Johns Hopkins University
- *edX Course: Robotics: Locomotion and Engineering* (Spring 2018), Penn Engineering Online Learning

Internship Mentor

- *Army Educational Outreach Program (AEOP) High School Internship Mentor* (Jun.–Aug. 2024), University of Pennsylvania

ADVISING AND MENTORING

Advising

- **Beining Qian** *Undergrad, Class 2023, Sichuan University*
  Undergrad Thesis: Loss shaping in flow matching
- **Shiyi Shen** *Ph.D. Student, Shanghai Jiao Tong University*
  Research: LoRA fine-tuning; Continual Learning
- **Hanwen Liu** *Ph.D. Student, Shanghai Jiao Tong University*
  Research: Efficient attention layers for LLMs
- **Yiwei Tan** *Master, Shanghai Jiao Tong University*
  Research: Operator Learning for solving PDEs
- **Qianjiang Yin** *Undergrad, Class 2023, Shanghai Jiao Tong University*
  Undergrad Thesis: Interpretable outlier sythesis
- **Shi Dong** *Undergrad, Class 2022, Shanghai Jiao Tong University*
  Undergrad Thesis: Understanding neural collapse in multi-layer ReLU networks

Mentoring

- **Fengrui Tian** *Ph.D. Student, University of Pennsylvania*
  Department of Computer and Information Science. Advisor: René Vidal
  Research Project: 3D dynamical scene generation
- **Prisha Shroff** *High School Intern, Hamilton High School*
  Army Educational Outreach Program (AEOP) High School Intern at University of Pennsylvania
  Research Project: Orthogonal matching pursuit for interpretable image classification
- **Salma Tarmoun** *Ph.D. Student, University of Pennsylvania*
  Department of Mathematics. Advisor: René Vidal
  Research Project: Gradient descent dynamics in attention models
- **Ziqing Xu** *Ph.D. Student, University of Pennsylvania*
  Wharton Statistics and Data Science. Advisor: René Vidal
  Research Projects: Convergence of gradient descent on linear networks; Convergence analysis of LoRA
- **Agustin Castellano** *Ph.D. Student, Johns Hopkins University*
  Department of Electrical and Computer Engineering. Advisor: Enrique Mallada
  Research Project: Reinforcement learning with almost sure safety

PROFESSIONAL SERVICES

Organizer

- Conference of Scientific Machine Learning (CSML), Shanghai, China, 2026
- IFAC World Congress Workshop on Contextual Control: Integrating Learning, Optimization and Physical Structures in Control Systems, Busan, Republic of Korea, 2026

Area Chair

- International Conference on Machine Learning (ICML), 2027
- International Conference on Learning Representations (ICLR), 2026, 2027
- International Conference on Artificial Intelligence and Statistics (AISTATS), 2026, 2027

Action Editor

- Journal of Machine Learning (JML)

Technical Reviewer

- *Journals*:
  IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI);
  Journal of Machine Learning Research (JMLR);
  Transactions on Machine Learning Research (TMLR);
  IEEE Transactions on Automatic Control (TAC);
  Automatica;
- *Conferences*:
  International Conference on Machine Learning (ICML), 2022-2026;
  Conference on Neural Information Processing Systems (NeurIPS), 2021-2025;
  International Conference on Learning Representations (ICLR), 2022-2025;
  International Conference on Artificial Intelligence and Statistics (AISTATS), 2025;
  IEEE Decision and Control (CDC), 2023,2025;
  American Control Conference (ACC), 2022;

PRESENTATIONS

- **Transformers Learn the Optimal DDPM Denoiser for Multi-Token GMMs**. Related work: [C23]
  Sep. 2026: Workshop on Frontiers in the Intersection of Stochastic Dynamical Systems and Machine Learning, Tianyuan Mathematics Research Center. Organizers: Tiejun Li, Xiang Zhou, Xiaoli Chen
- **Learning Dynamics in the Feature-Learning Regime: Implicit Bias, Neural Collapse and Optimal Denoising**. Related work: [C23, C19, C17, C14, C12]
  Aug. 2026: BIRS Workshop on Mathematical Foundations in Deep Learning and Generative AI. Organizers: Qing Qu, et al.
  Jul. 2026: IoT Thrust Seminar, Hong Kong University of Science and Technology (Guangzhou). Host: Zhengyi Yuan
  Jul. 2026: Department of Computer and Information Science, University of Macau. Host: Peng Wang
- **Slow Coherency, Aggregation and Clustering in Networked Systems**. Related work: [J5, C9, C8, C3]
  May. 2026: Green Control Workshop 2026, Peking University. Host: Pengcheng You
- **Understanding Incremental Learning with Closed-form Solution to Gradient Flow on Overparametrized Matrix Factorization**. Related work: [C20]
  Dec. 2025: 64th IEEE Conference on Decision and Control, Rio de Janeiro, Brazil
- **Learning Dynamics in the Feature Learning Regime: Implicit Bias, Neural Collapse, and Robustness**. Related work: [C19, C17, C14, C12]
  Sep. 2026: Minisymposium on Mathematical Theory of Deep Learning, Shanghai SIAM 2026. Organizer: Shuai Lu, Lei Shi.
  Nov. 2025: Workshop on Recent Advances in Optimization, Control and AI, New York University Shanghai, Shanghai. Hosts: Shuyang Ling, Mathieu Lauriere
- **Simplicity Bias in Shallow Neural Networks**. Related work: [C17, C14, C12]
  Jul. 2025: School of Data Science, The Chinese University of Hong Kong, Shenzhen. Host: Yan Jiang, Tongxin Li
  Jan. 2025: School of Data Science, The Chinese University of Hong Kong, Shenzhen
  Jan. 2025: Institute of Natural Sciences, Shanghai Jiao Tong University
  Oct. 2024: "AI + MATH" Colloquia, Virtual, Shanghai Jiao Tong University. Host: Zhiqin Xu
- **Early Neuron Alignment in Two-layer ReLU Networks with Small Initialization**. Related work: [C12]
  Nov. 2023: DeepMath 2023, Johns Hopkins University
- **Spectral Clustering and Model Reduction for Weakly-Connected Coherent Network Systems**. Related work: [C8]
  Jun. 2023: American Control Conference 2023, San Diego, CA
- **Exploiting Structural Properties in the Analysis of High-dimensional Dynamical Systems**. Related work: [C12, C10, C9, C5]
  Jan. 2023: University of Michigan. Host: Necmiye Ozay
- **Learning Coherent Clusters in Weakly-Connected Network Systems**. Related work: [C9]
  Jan. 2023: ROSEI Summit, Johns Hopkins University
- **Convergence and Implicit Bias of Gradient Flow on Overparametrized Linear Networks**. Related work: [P1, C5]
  Jun. 2022: RSRG Seminar, California Institute of Technology. Hosts: Adam Wierman, Steven Low
  Jun. 2022: Semiautonomous seminar, UC Berkeley. Hosts: Chinmay Maheshwari, Shankar Sastry
  Mar. 2022: MINDS Retreat, Johns Hopkins University
  Jan. 2022: 2022 TRIPODS Winter School on Interplay between AI and Dyn. Sys., Virtual
- **Accurate Reduced Order Models for Coherent Heterogeneous Generators**. Related work: [J1, C4]
  May. 2021: American Control Conference 2021, Virtual
- **Dynamics Concentration of Tightly-Connected Large-Scale Networks**. Related work: [C3]
  Dec. 2019: 58th IEEE Conference on Decision and Control, Nice, France

REFERENCES

Enrique Mallada *Ph.D. Advisor*
Associate Professor, Electrical and Computer Engineering
Johns Hopkins University, Baltimore, MD

René Vidal *Postdoc Advisor*
Rachleff University Professor, Electrical and Systems Engineering
University of Pennsylvania, Philadelphia, PA

Fernando Paganini
Professor, Electrical and Telecommunications Engineering
Universidad ORT Uruguay, Montevideo, Uruguay

Eduardo Sontag
University Distinguished Professor, Electrical and Computer Engineering
Northeastern University, Boston, MA

Hamed Hassani
Professor, Electrical and Systems Engineering
University of Pennsylvania, Philadelphia, PA

Necmiye Ozay
Chen-Luan Family Faculty Development Professor, Electrical and Computer Engineering
University of Michigan, Ann Arbor, MI